spread to our villages. The ICMR study showed that if the prevalence of diabetes stood at 16.4 per cent of the population in urban India, it was also 8.9 per cent in rural India. Another ICMR study published in the journal *Diabetologia* in 2011 had found this figure to be 2.7 per cent in rural India. "We are certainly seeing this trend. Cases from rural areas are rising. People come to cities and pick up the urban lifestyle—packaged, convenience food and less exercise. They then take this habit back with them," says Dr Dheeraj Kapoor, head of endocrinology at the Kokilaben Dhirubhai Ambani Hospital in Mumbai. This is what gave 38-year-old Shibani Kapur's younger, unemployed brother diabetes. For, every time the Noida-based customer care executive visited her village Ajnauli, near Meerut, she carried boxes of biscuits, sweets, chips and soda. Two years ago, he was diagnosed with diabetes. "Nobody in our family has ever had it," she says. "Now he has to come to Delhi to get affordable medicine and treatment. Doctors say he has to exercise and cut out all sugar."

A WEIGHTY LIFESTYLE PROBLEM

Due to our long, primordial state as hunter-gatherers, when supply of food was uncertain, the human body evolved a mechanism to save energy from food, through fat, as a reserve during possible scarcity. In our era of mechanised agriculture and refrigeration, humans not only eat well and more, but this store of fat in their bodies is now a main reason for insulin resistance.

The impact of fat on insulin resistance is well researched. A 2020 study in the journal *Plos One* found that an increase in BMI (body mass index) increases the risk of diabetes and pre-diabetes, overweight individuals being at a decidedly higher risk. Another 2020 study in *Diabetologia* found obesity increases the risk of Type 2 diabetes by at least six times. How does this happen? Abdominal fat usually releases pro-inflammatory chemicals which can make the body less sensitive to the insulin it produces. Too much fat can also cause the pancreas to oversecrete insulin, the consistently high levels becoming another cause for resistance to it. In addition, a diet loaded in simple sugars (food that lacks fibre and whole grain) is quickly digested, leading to a spike in insulin. "Inefficient insulin, an outcome of too much fat in the belly, and reduced storage of excess glucose due to loss of muscle [as a result of

GASTRONOMIC SHIELD AGAINST DIABETES

Calories and carbs, fibre and fat—every meal needs watching to keep diabetes at bay. **Dr Ritika Samaddar**, regional head, Nutrition and Dietetics, Max Super Speciality Hospital, Delhi, tells us how



BEWARE THINGS ENDING WITH 'OSE'

Fructose, glucose, sucrose, maltrose—all signs that your product has added sugar. Also, stay clear of 'confectionery sugar', 'cane sugar' and 'HFCS' (high fructose corn syrup). Check the labels. Added sugars are a cause of diabetes and obesity; it is important to keep them under limit



NOT EVERYTHING 'HEALTHY' IS HEALTHY

Added sugars are present in unforeseen places: salad dressings, yoghurt, breakfast cereals



CONSUME FRESH FRUIT

The quantity of fruit intake can be increased for diabetics. In addition to natural sugar, fruits have high fibre, antioxidants and micronutrients. Fresh fruit is better than packaged food with hidden sugars to satisfy hunger cravings



CUT OUT THE REFINED CARBS

The glycemic index of refined carbohydrates is not much better than sugar. Items made out of refined grain like breads and croissants can spike blood sugar. These should be restricted or replaced with items made of whole grains and high in fibre



NOT SUGAR ALTERNATIVES

Many people replace white refined sugar with honey, jaggery or brown sugar. The glycemic index of these are the same as sugar. One can't have them in unlimited amounts. Artificial sweeteners should also be taken in minimal amounts



BREAK FAST WISELY

Fasting can help improve the action of insulin, and is helpful for diabetics facing insulin resistance. But it's important to break the fast with a complex carb, for release of sugar shouldn't be instant. A fruit/dried fruit is better than juice